

Flutter Development Interns Task

Week 1: Basic Flutter Development and UI Building

Learning Objectives:

- Understand Flutter's basic structure.
- Build simple and responsive user interfaces.
- Learn navigation between screens.

Tasks:

1. Set Up Environment:

Install Flutter and set up Android Studio/VS Code.

- Create a simple Flutter project.

2. Create a Basic UI:

Build a login screen with two `TextFormField`s for email and

password.

- Add a `FlatButton` for login and a `Text` widget for "Forgot

Password?"

- Use basic widgets like `Column`, `Row`, and `Container` to structure the UI.

3. Navigation Between Screens:

Create a second screen (e.g., Home Screen).

- Implement navigation from the login screen to the home screen using `Navigator.push()`.

4. Form Validation:

Validate the email field for proper email format.

- Ensure the password field is not empty.

Deliverables:

- A basic app with a login screen and navigation to the home screen.

- GitHub repository with the code and a simple README

Week 2: Data Management and Persistent Storage

Learning Objectives:

- Understand basic state management.
- Save and retrieve data locally.

Tasks:

1. State Management Basics:

Use `setState` to manage the state of widgets.

- Create a simple counter app where the user can increase or decrease a counter value.

2. Persistent Data Storage:

Use `SharedPreferences` to save and retrieve data.

- Add functionality to save the counter value and retrieve it on app restart.

3. Simple List App:

Create a to-do list app with the following features:

- Add tasks.
- Display tasks in a `ListView`.
- Save tasks using `SharedPreferences`.

Deliverables:

- A to-do list app with local storage functionality.
- GitHub repository with the code and a README explaining the app.

Week 3: Finishing Touches and Final Project

Learning Objectives:

- Combine learned concepts into a small functional app.

- Test and debug the app.

Tasks:

1. Final Project: Task Management App

Create a simple app with:

- A home screen displaying a list of tasks.
- Functionality to add, delete, and mark tasks as complete.
- Data persistence using [SharedPreferences](#).

2. Testing and Debugging:

Test the app for proper navigation and data saving.

- Debug any issues using Flutter's built-in debugging tools.

3. UI Enhancements:

Add a custom app bar with a title and an action button (e.g., for adding a task).

- Use basic icons from the [Icons](#) library for visual appeal.

Deliverables:

- A functional task management app.

GitHub repository with code and a README including setup instructions.

A short video demonstrating the app.

Deadline: 26th June, 2026

Additional Notes:

- **Bonus Challenge:** For interns wanting to explore more, they can optionally:
 - Add a splash screen.
 - Explore Firebase integration (optional and not required for beginners).